HDOC Day-47

Aim: Write a C program using pointers and a separate function **ceil_index()** to find the smallest element greater than or equal to the target in a sorted array. If the value repeats, display its first index. Print -1 if it does not exist.

C Program:

```
#include <stdio.h>

void ceil_index(int *arr, int n, int target)
{
    int i;
    for (i = 0; i < n; i++)
    {
        if (*(arr + i) >= target)
        {
            printf("Ceil of %d: %d at index %d\n",
                   target, *(arr + i), i);
            return;
        }
    }
    printf("-1\n");
}

int main()
{
    int n, i, target;
    printf("Enter size of array: ");
    scanf("%d", &n);
    if (n <= 0)
    {
        printf("Invalid array size\n");
        return 0;
    }
    int arr[n];
    printf("Enter sorted array elements: ");
    for (i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    printf("Enter target: ");
    scanf("%d", &target);
    ceil_index(arr, n, target);
    return 0;
}
```

Sample Outputs:

Example 1

```
Array: 5 7 7 8 8 10
Target: 6
Output: Ceil of 6: 7 at index 1
```

Example 2

```
Array: 5 7 7 8 9 10
Target: 8
Output: Ceil of 8: 8 at index 3
```

Example 3

```
Array: 5 7 7 8 8 10
Target: 11
Output: -1
```